

Experimental Setup

Does SS OPS 1 conserve its reciprocal event ledger and respect causal rescheduling?

Experiment: gross.reciprocal-events.v1 Model: signal-space.ss-ops-1.v1
Runtime configuration: fixtures/research/gross-test-02.json
Plan lock: fa05c1715458702d004240a27a60d9313427db2061a40b98037e7cbe66bd7c4f

MODEL AND NUMERICAL SETUP

Action: $H = \kappa d \text{-dagger} P d + \lambda N^2/2$; $d = z_1 - z_2$, $P = ww \text{-dagger}$, $N = |z_1|^2 + |z_2|^2$; canonical complex Hamiltonian flow, no frozen memory in primary model.
Units: All states, couplings and gate parameter s dimensionless. A gate spans $s=0$ to 1; s is not proper time.
Domain: 12 directed events; 8 complex wave wires, 4 unit memory wires; explicit ordered reuse, identity physical transport. Four preparations at $\lambda=0$ and 0.1, $\kappa=\pi/2$.
Boundaries: Finite acyclic incidence, including a disconnected branch. No spatial mesh, boundary condition or continuum limit. Complete-cut charge accounts for every wire; event H is not asserted to be circuit energy.
Initial Data: PCG64 runtime initialization stream from root 2026092502; normalized independent complex normal directions; wave norms uniform $[0.35, 0.85]$, memories unit norm; independent $U(2)$ event frames.
Solver: DOP853 $\text{rtol}=2e-12$ $\text{atol}=2e-14$ $\text{max_step}=0.125$, repeated at half both tolerances. Nine saved samples per gate. Exact constant- S solution is an independent control, not the numerical solver.
Tolerances: Residuals $<1e-9$. Controls $>1e-4$ in every preparation. Complete real input Jacobian on preparation 0 for both λ s; memory tangent directions preserve unit norm; central steps $1e-5$ and $5e-6$, agreement $<1e-6$.

COMPETING PREDICTIONS

reciprocal-local: Conservation and permissible schedules pass; no outside-ancestor response; independent frames agree; physical interventions are detected.
inconsistent: Missing backreaction, forbidden causal response or dependence on arbitrary event scheduling exceeds the frozen tolerances.

CONTROLS

exact-and-refined: Discriminate integration error from event-law failure; method: Compare all saved circuit trajectories to closed constant- S solution; halve both numerical tolerances.
causal-and-frame: Distinguish physical routing from description; method: Compare two topological schedules and independent $U(2)$ event frames with transformed wire transports; full central-difference input directions at two steps.
physical-interventions: Expose noncommutativity and missing reciprocal response; method: Reverse two otherwise disjoint wave-pair events sharing one memory; separately freeze driven memory and compare with its exact external-memory solution.

PREREGISTERED DECISIONS

coverage (All 4 preparations, both λ s, 12 events and all declared comparisons are saved)
Pass: All 4 preparations, both λ s, 12 events and all declared comparisons are saved
Fail: Complete evidence violates the declared threshold.
Unresolved: Incomplete evidence or numerical failure; no pass.
Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

conservation (N , memory norm, J , event H , projector and complete-cut charge residuals below $1e-9$ at both tolerances)

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Pass: N, memory norm, J, event H, projector and complete-cut charge residuals below $1e-9$ at both tolerances

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

exact-reference (Reciprocal and frozen numerical trajectories agree with independent closed solutions below $1e-9$)

Pass: Reciprocal and frozen numerical trajectories agree with independent closed solutions below $1e-9$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

tolerance-refinement (Halving $rtol$ and $atol$ changes the circuit state and overlap records by less than $1e-9$)

Pass: Halving $rtol$ and $atol$ changes the circuit state and overlap records by less than $1e-9$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

rescheduling (Permissible schedules agree on states and event overlap records below $1e-9$)

Pass: Permissible schedules agree on states and event overlap records below $1e-9$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

local-frames (Independent $U(2)$ event frames with transformed wire transport agree below $1e-9$)

Pass: Independent $U(2)$ event frames with transformed wire transport agree below $1e-9$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

causal-jacobian (Outside-ancestor response below $1e-9$; nonzero inside response above $1e-4$ at both difference steps)

Pass: Outside-ancestor response below $1e-9$; nonzero inside response above $1e-4$ at both difference steps

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

jacobian-refinement (Halving the central-difference step changes the Jacobian by less than $1e-6$)

Pass: Halving the central-difference step changes the Jacobian by less than $1e-6$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error

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bounds, not statistical confidence.

shared-memory-order (Reversing the shared-memory control changes its final projector by more than $1e-4$ in each preparation)

Pass: Reversing the shared-memory control changes its final projector by more than $1e-4$ in each preparation

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

frozen-memory (Freezing driven memory exposes matrix-charge error above $1e-4$; reciprocal memory responds above $1e-4$)

Pass: Freezing driven memory exposes matrix-charge error above $1e-4$; reciprocal memory responds above $1e-4$

Fail: Complete evidence violates the declared threshold.

Unresolved: Incomplete evidence or numerical failure; no pass.

Uncertainty: Tolerance halving, exact solution and finite-difference step comparison; engineering error bounds, not statistical confidence.

VISUAL QUESTIONS

causal-incidence [evidentiary]: Do event outputs depend only on their causal ancestors?

View: Directed incidence and measured Jacobian heatmap; nonancestors remain visible.; observables: response, ancestor

reciprocal-local: Residuals below locked tolerances; causal support follows incidence; physical controls remain nonzero.

inconsistent: Forbidden response, ledger defect, description dependence or undetected intervention.

Controls: exact-and-refined, causal-and-frame, physical-interventions; uncertainty: Numerical refinement and exact reference; finite preparations. Zeros retained; no smoothing.

Data: reports/<report-id>/plot-data/incidence.json, reports/<report-id>/plot-data/jacobian.csv;

transforms: No fitting or downsampling; log values display $\max(\text{value}, 1e-18)$, exact zeros retained;

Jacobian uses central differences and norm across output participants.

matrix-balance [evidentiary]: Does reciprocal memory exchange close the matrix-charge ledger?

View: Signed wave/memory exchange and reciprocal versus frozen total drift through a gate.; observables: wave_z_change, memory_z_change, total_change

reciprocal-local: Residuals below locked tolerances; causal support follows incidence; physical controls remain nonzero.

inconsistent: Forbidden response, ledger defect, description dependence or undetected intervention.

Controls: exact-and-refined, causal-and-frame, physical-interventions; uncertainty: Numerical refinement and exact reference; finite preparations. Zeros retained; no smoothing.

Data: reports/<report-id>/plot-data/balance.csv; transforms: No fitting or downsampling; log values

display $\max(\text{value}, 1e-18)$, exact zeros retained; Jacobian uses central differences and norm across output participants.

numerical-audit [evidentiary]: Do conservation and equivalent descriptions survive numerical refinement?

View: Baseline/refined conservation residuals and independent state/record equivalence checks.;

observables: N_error, memory_norm_error, J_error, H_error, exact, refinement, rescheduling, frame

reciprocal-local: Residuals below locked tolerances; causal support follows incidence; physical controls remain nonzero.

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inconsistent: Forbidden response, ledger defect, description dependence or undetected intervention.
Controls: exact-and-refined, causal-and-frame, physical-interventions; uncertainty: Numerical refinement and exact reference; finite preparations. Zeros retained; no smoothing.
Data: reports/<report-id>/plot-data/residuals.csv, reports/<report-id>/plot-data/summary.json;
transforms: No fitting or downsampling; log values display $\max(\text{value}, 1e-18)$, exact zeros retained;
Jacobian uses central differences and norm across output participants.

order-controls [evidentiary]: Can the audit distinguish physical interventions from rescheduling?
View: Control response for every preparation and both lambdas with fixed detection threshold.;
observables: order_signal, frozen_charge, memory_response
reciprocal-local: Residuals below locked tolerances; causal support follows incidence; physical controls remain nonzero.
inconsistent: Forbidden response, ledger defect, description dependence or undetected intervention.
Controls: exact-and-refined, causal-and-frame, physical-interventions; uncertainty: Numerical refinement and exact reference; finite preparations. Zeros retained; no smoothing.
Data: reports/<report-id>/plot-data/controls.csv; transforms: No fitting or downsampling; log values display $\max(\text{value}, 1e-18)$, exact zeros retained; Jacobian uses central differences and norm across output participants.

RESOURCE LIMITS

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{"max_memory_mb": 512, "max_output_mb": 32, "max_wall_seconds": 120}
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PROHIBITED INFERENCES

No metric, propagation spectrum, clock, mechanical recoil or energy-momentum interpretation follows from this event audit.

Finite-difference causal checks and four preparations are not a global proof. Local $U(2)$ covariance is not Lorentz covariance.

Drift, anisotropy and all-sector propagation are not evaluated here; keep Tests 3,4,11 open.